TARUN SINGH

PhD Scholar | Department of Civil Engineering | IIT Roorkee

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RESEARCH PROFILE

PhD researcher working on constitutive modelling of rate-dependent materials, with emphasis on finite-deformation frameworks, rheological connections, evolving natural configurations, and thermodynamically consistent modelling of viscoelastic solids.

EDUCATION

PhD, Civil Engineering | 2023–Present
Indian Institute of Technology Roorkee
Research: Constitutive Modelling of Rate-Dependent Materials | Supervisor: Dr. Sandipan Paul

M.Tech., Structural Engineering | 2020–2022
National Institute of Technology Jalandhar | Supervisor: Dr. Navdeep Singh
Thesis: Carbonation Resistance of Concrete Made with Iron Slag and Recycled Concrete Aggregate Dr. Sandeep Chaudhary

B.Tech., Civil Engineering | 2015–2019
Rewa Engineering College, Rewa, Madhya Pradesh | Supervisor: Dr. Sandeep Choudhary
Project: Development of Porous Asphalt Pavement

RESEARCH EXPERIENCE

- PhD research on constitutive modelling of rate-dependent materials and finite-deformation viscoelasticity.
- Development of constitutive frameworks using rheological connections and multiple/evolving natural configurations.
- Formulation and numerical integration of nonlinear constitutive equations, with parameter identification using experimental response data.
- M.Tech. research on durability and performance of concrete incorporating iron slag and recycled concrete aggregates.

SELECTED PUBLICATIONS

Singh, T. and Paul, S. (2026). "Constitutive modeling of Kelvin–Voigt type viscoelastic solids undergoing finite deformations using the theory of evolving natural configurations." *International Journal of Solids and Structures, 341, Article 114336.* DOI: https://doi.org/10.1016/j.ijsolstr.2026.114336

Singh, T. and Paul, S. (2025). "On the extension of the concept of rheological connections to a finite deformation framework using multiple natural configurations." *International Journal of Engineering Science, 212, Article 104267.* DOI: https://doi.org/10.1016/j.ijengsci.2025.104267

Singh, N., Singh, A., Ankur, N., Kumar, P., Kumar, M., and Singh, T. (2022). "Reviewing the properties of recycled concrete aggregates and iron slag in concrete." *Journal of Building Engineering, 60, 105150.*

Singh, N., Ankur, N., Singh, T., Kumar, M., Singh, A., and Kumar, P. (2022). "Non-destructive evaluation of iron slag concrete containing recycled concrete aggregates." *Indian Concrete Journal, 96(11).*

Singh, N., Singh, T., Kumar, M., Singh, A., and Kumar, P. (2022). "Investigating the fresh state performance of concrete containing iron slag and recycled concrete aggregates." *Materials Today: Proceedings, 65, 1467–1477.*

CONFERENCE PRESENTATIONS

Singh, T. and Paul, S. (2026). "Thermodynamically Consistent Constitutive Model for Viscoelastic Solids Under Finite Deformation Using Evolving Natural Configurations." *7th Indian Conference on Applied Mechanics, IIT Kanpur.*

Singh, T. and Paul, S. (2025). "Challenges in extending rheological connections to a finite deformation theory." *14th International Symposium on Plasticity and Impact Mechanics (IMPLAST 2025), IIT Roorkee.*

RESEARCH INTERESTS

Constitutive modelling of rate-dependent materials • Finite-deformation continuum mechanics • Finite-strain viscoelasticity and viscoplasticity • Rheological models and connections • Evolving and multiple natural configurations • Computational mechanics and finite-element implementation • Sustainable construction materials

TECHNICAL SKILLS

Scientific computing: MATLAB, Python

Finite-element and constitutive implementation: Abaqus, UMAT, Fortran, FEniCSx

Numerical methods: Numerical integration of constitutive equations, parameter identification and model calibration